

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_12apr25_isolado.parquet | Experiment: rssi_odom_dataset_11apr25_isolado.parquet

model_type: GRU
group_sec: 100
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 256
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 50
hyperband_iterations: 3
group_msec: 100
samples_per_sec: 10
forecast_horizon_sec: 10
forecast_steps: 100
timesteps: 60

Dataset Parameters:

Training: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60

Experiment: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60

Tuning:

Trials executed: 10/10

Training Performance:

Loss: 0.0077 | MAE: 0.0668

Validation Performance:

Loss: 0.0183 | MAE: 0.1117

GRU Multi-Step Training Report

Report Generated: 2025-08-05 16:50:52

Training Start: 2025-08-05_15-33-13

Training End: 2025-08-05_15-41-45

Training Duration: 8.54 minutes

Predictions Start: 2025-08-05_15-41-45

Predictions End: 2025-08-05_15-46-35

Predictions Duration: 4.84 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.8077

MSE: 5.5105

RMSE: 2.3474

R2 Score: 0.7883

Dataset size before down-sample: 40,318 rows
Dataset size after down-sample: 4,847 rows

First 5 timestamps BEFORE:

1744397593300
1744397593310
1744397593332
1744397593362
1744397593375

...

Last 5 timestamps BEFORE:

1744398077967
1744398077978
1744398077991
1744398078003
1744398078016

First 5 timestamps AFTER:

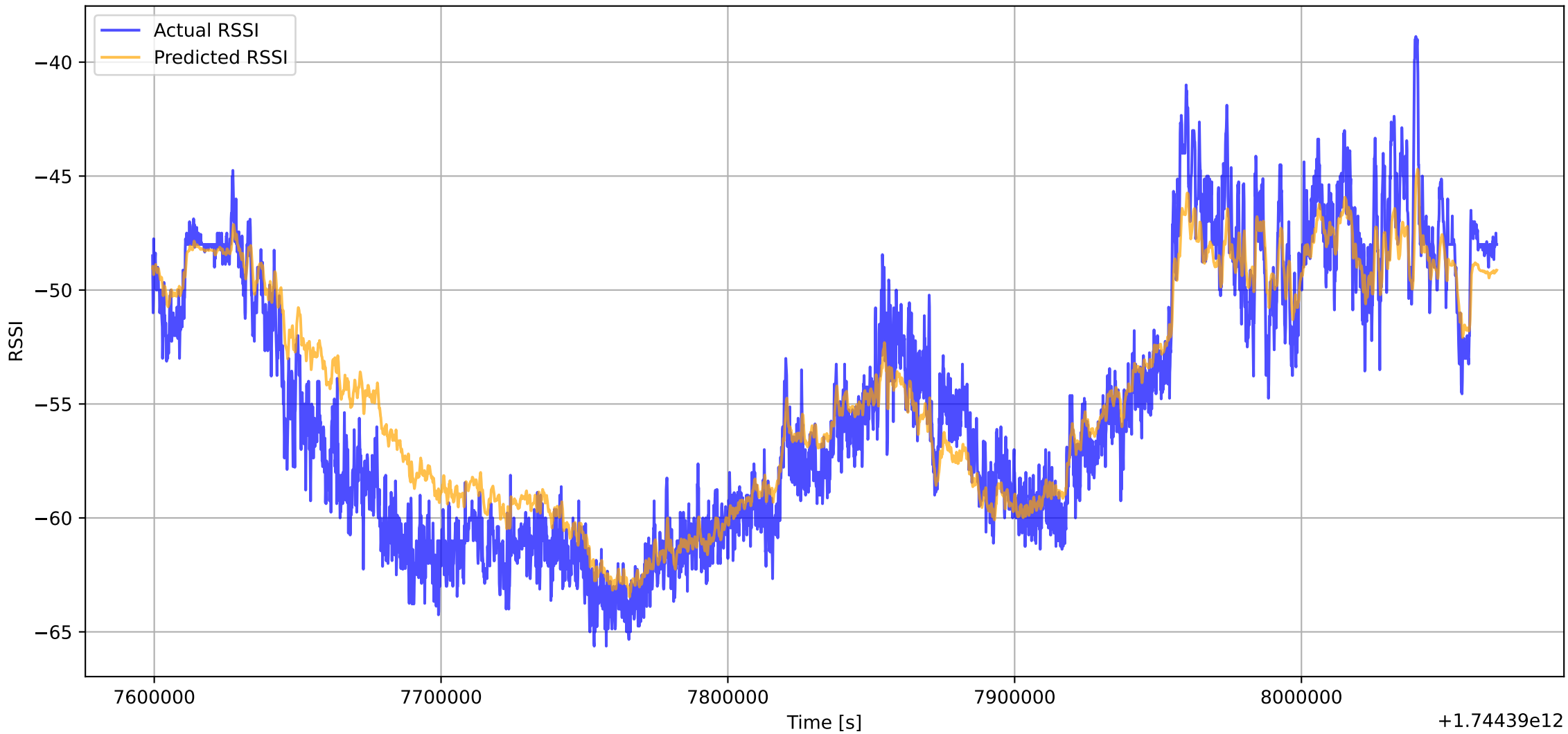
1744397593300
1744397593536
1744397593601
1744397593706
1744397593807

...

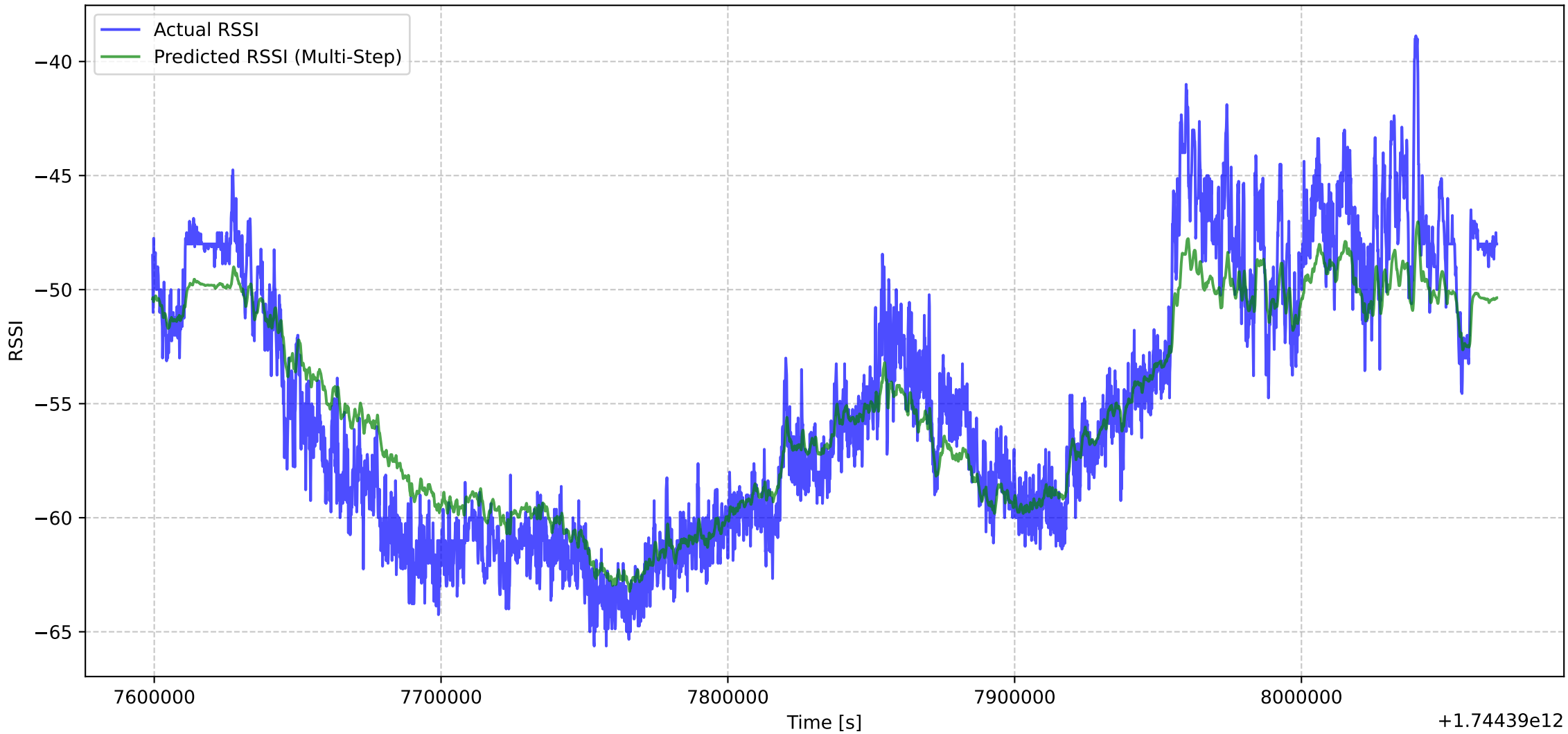
Last 5 timestamps AFTER:

1744398077608
1744398077705
1744398077811
1744398077907
1744398078003

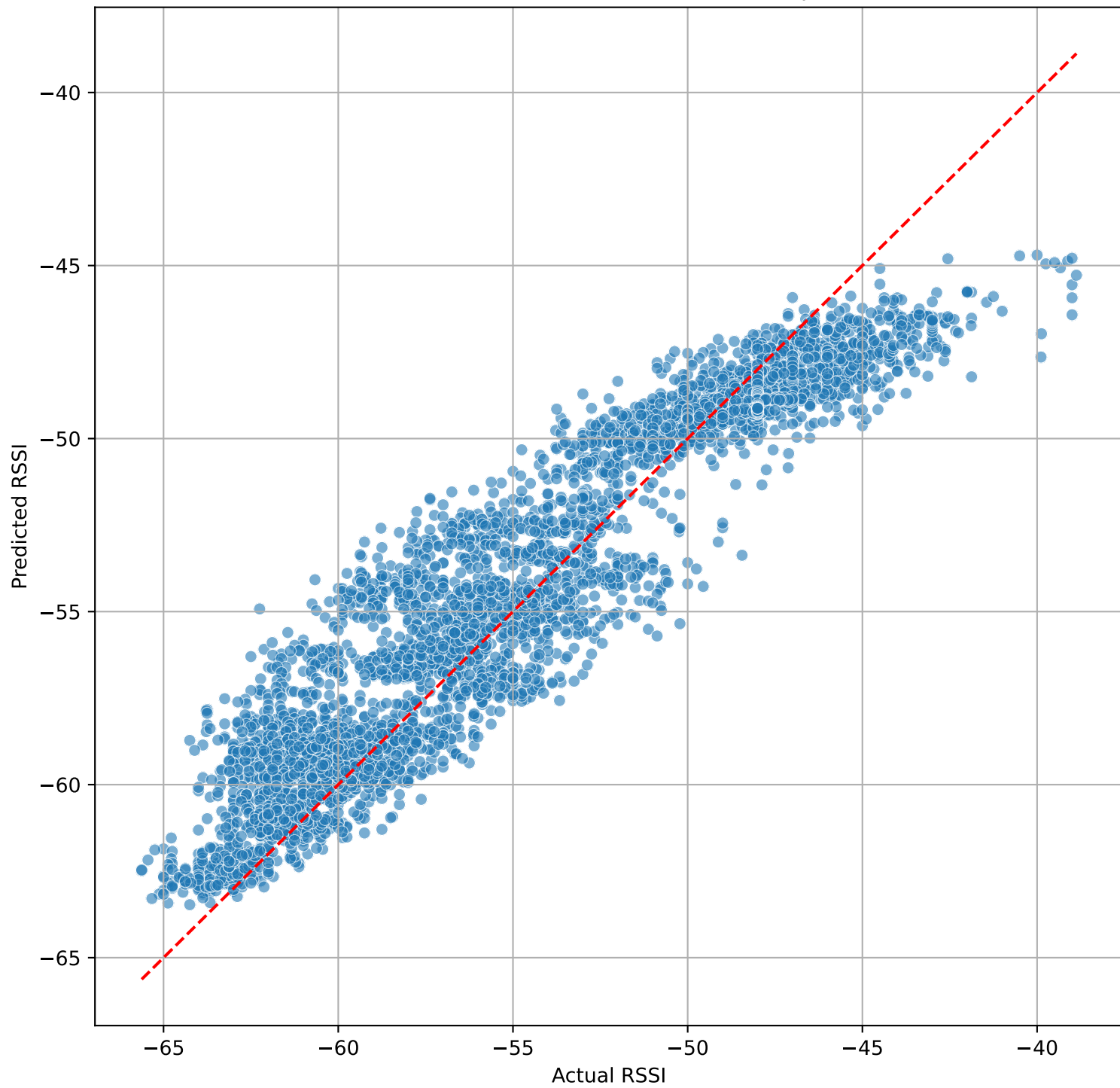
Actual vs Predicted RSSI (First Step)



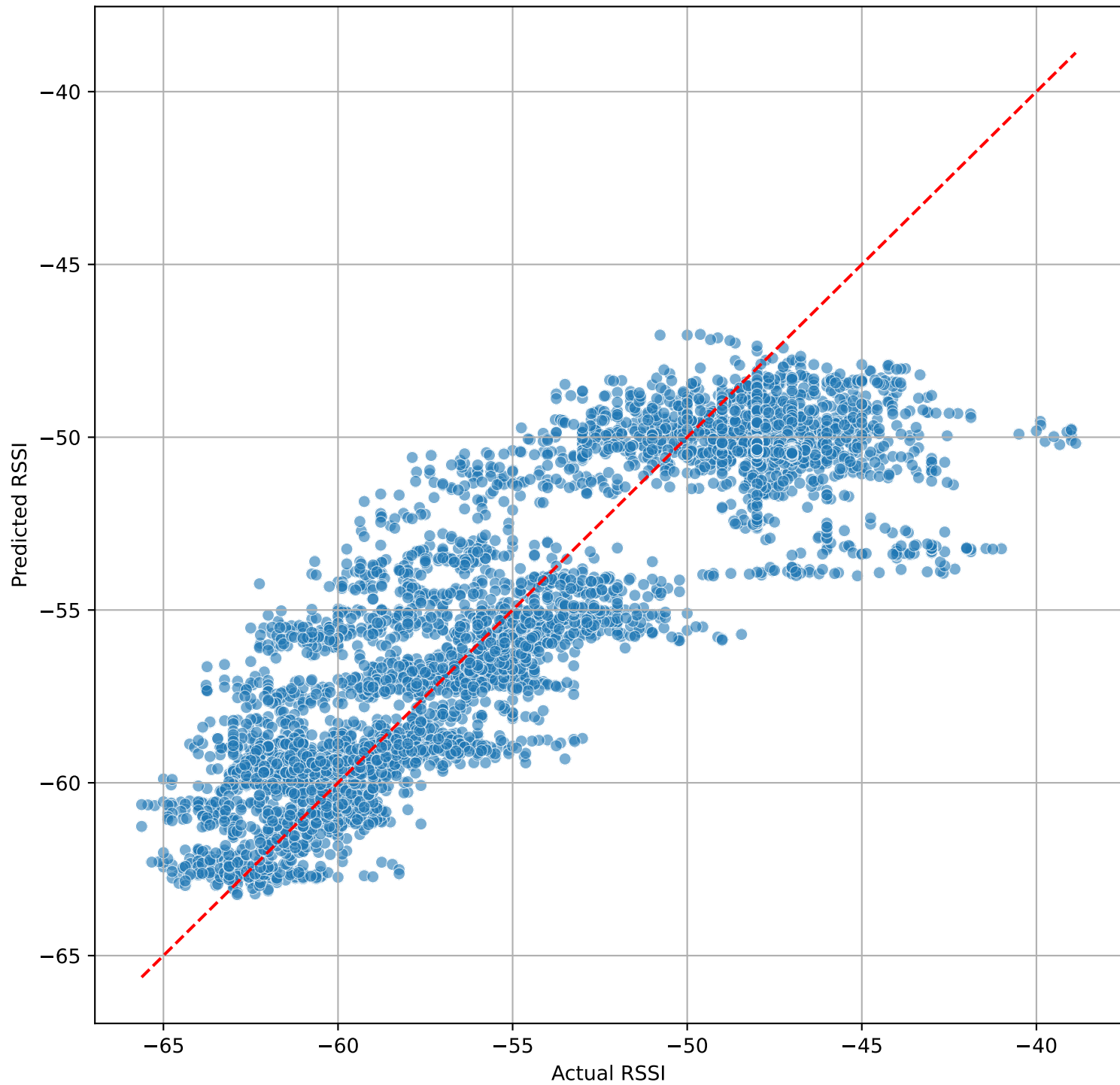
Actual vs Predicted RSSI (Multi-Step (step 100 $\approx$ 10.0 s))



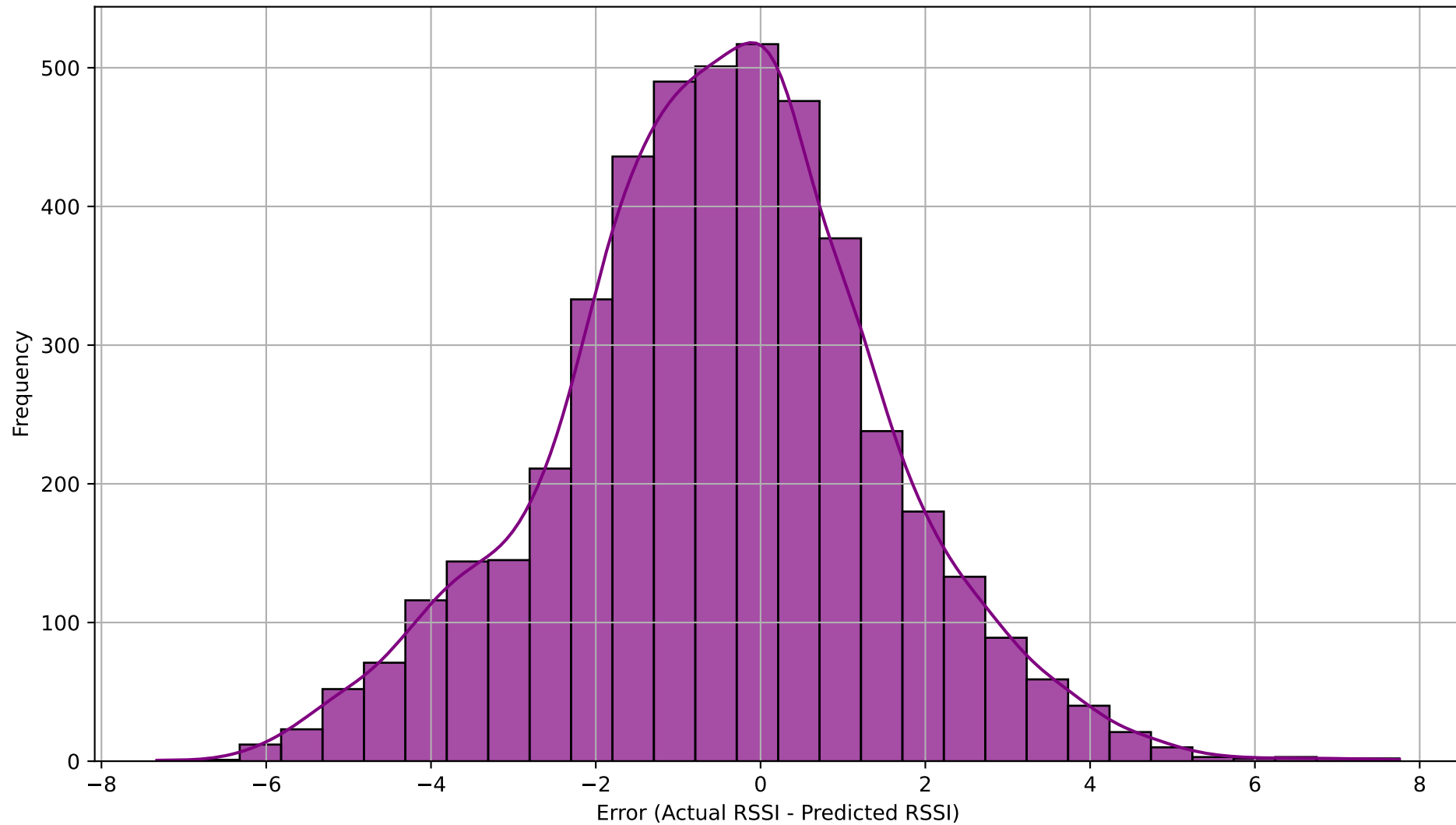
Actual vs Predicted RSSI (First Step)



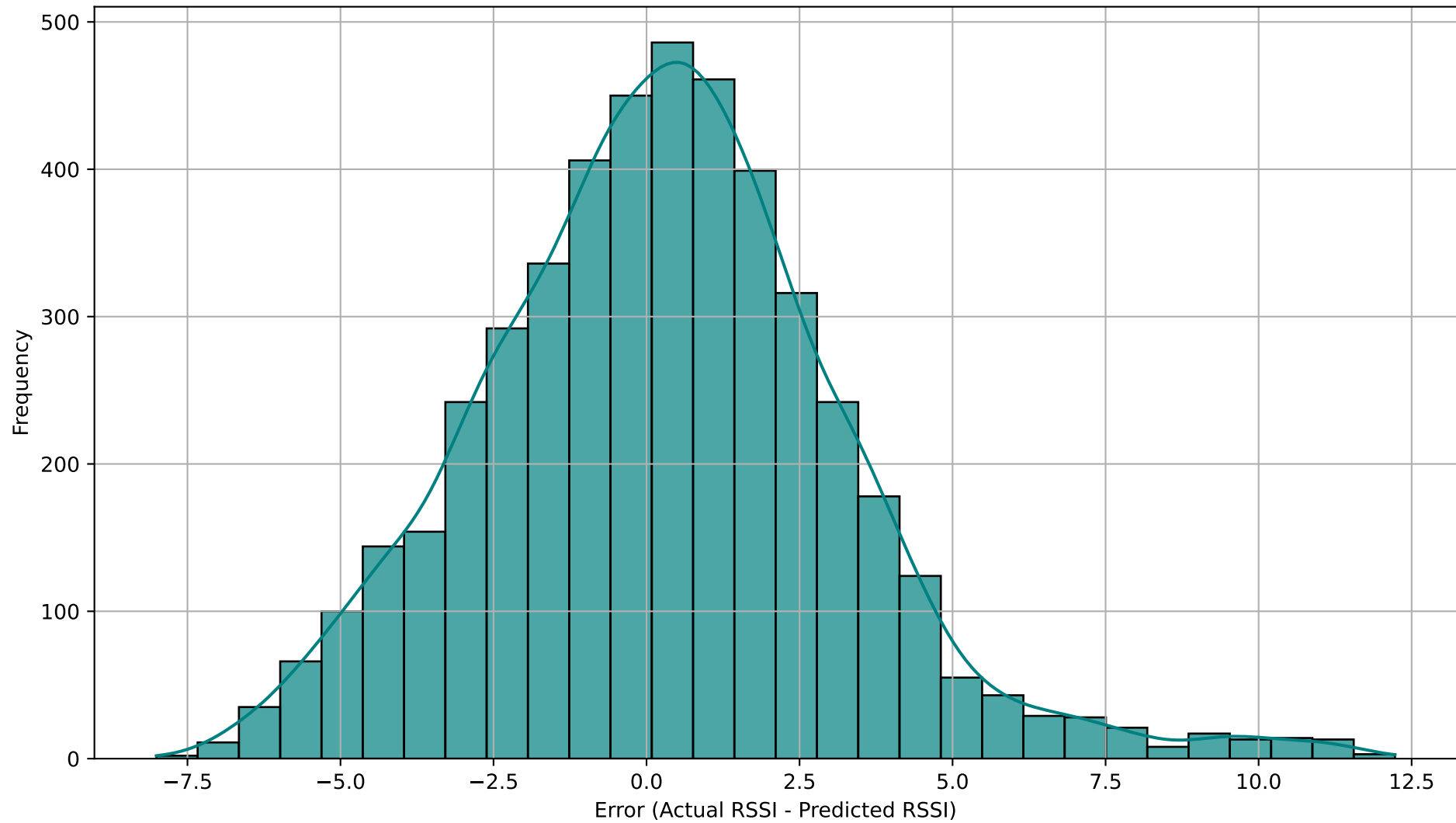
Actual vs Predicted RSSI (Last Step - step 100 $\approx$ 10.0 s)



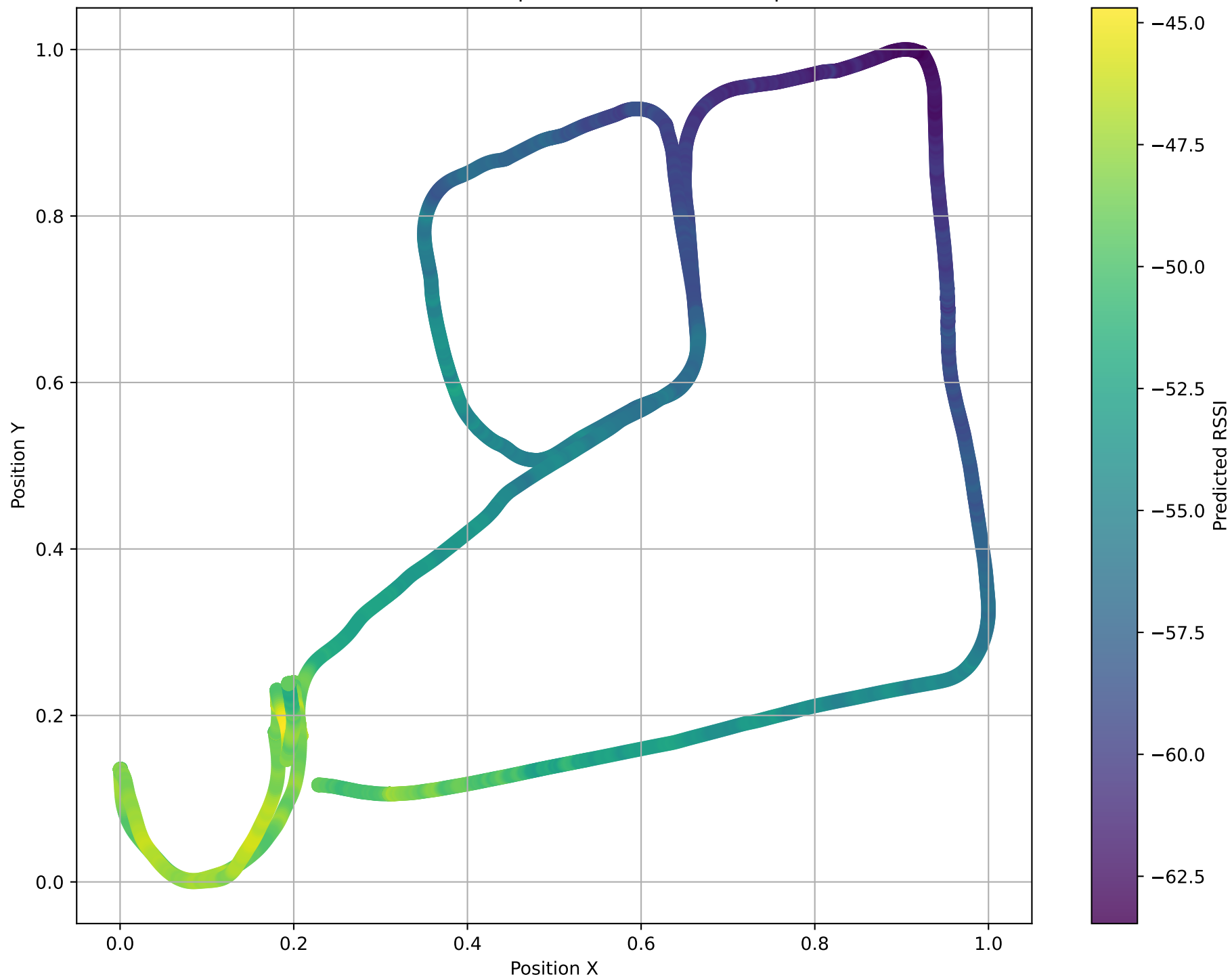
Distribution of Prediction Errors (First Step)



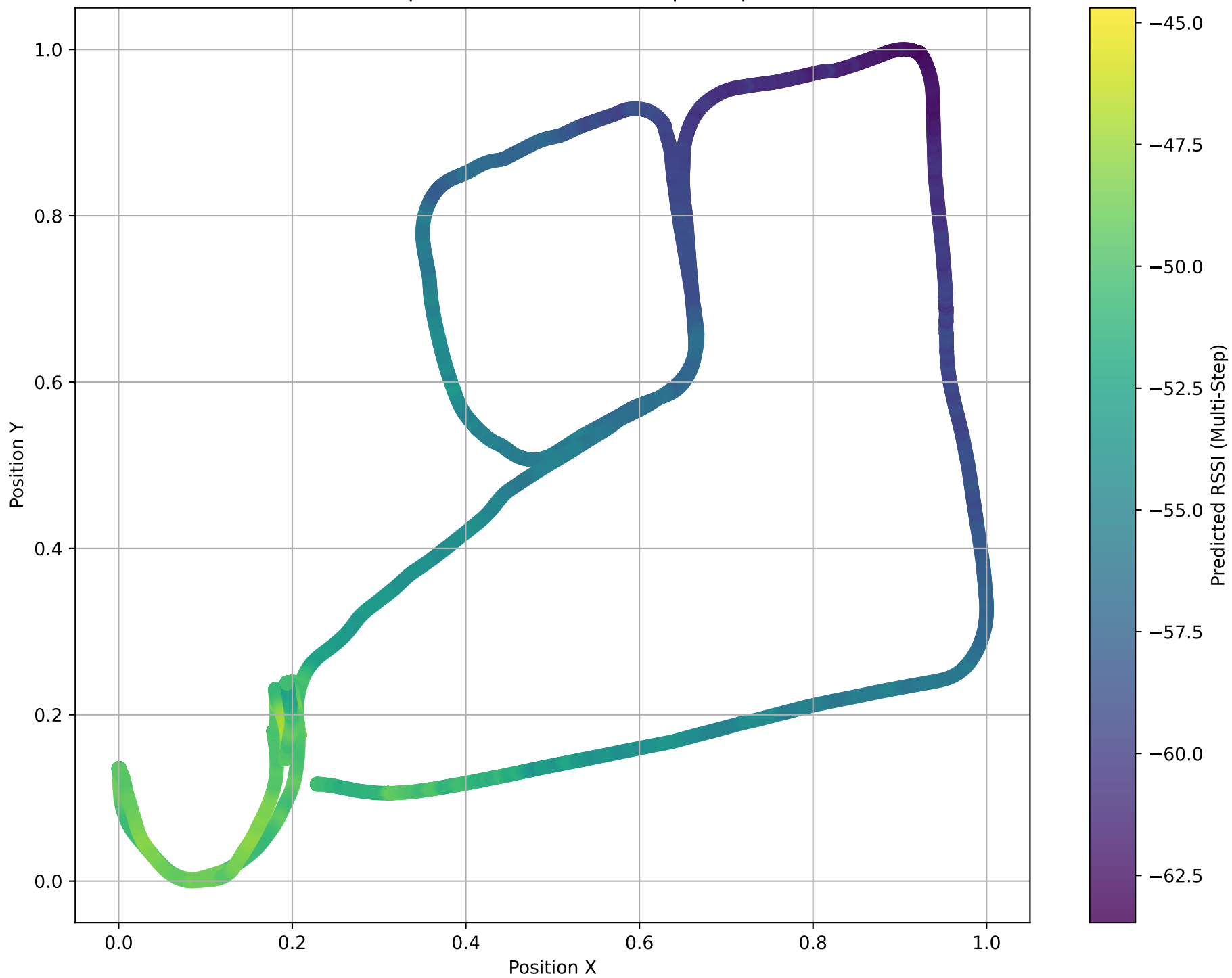
Distribution of Prediction Errors (Last Step - step 100 $\approx$ 10.0 s)



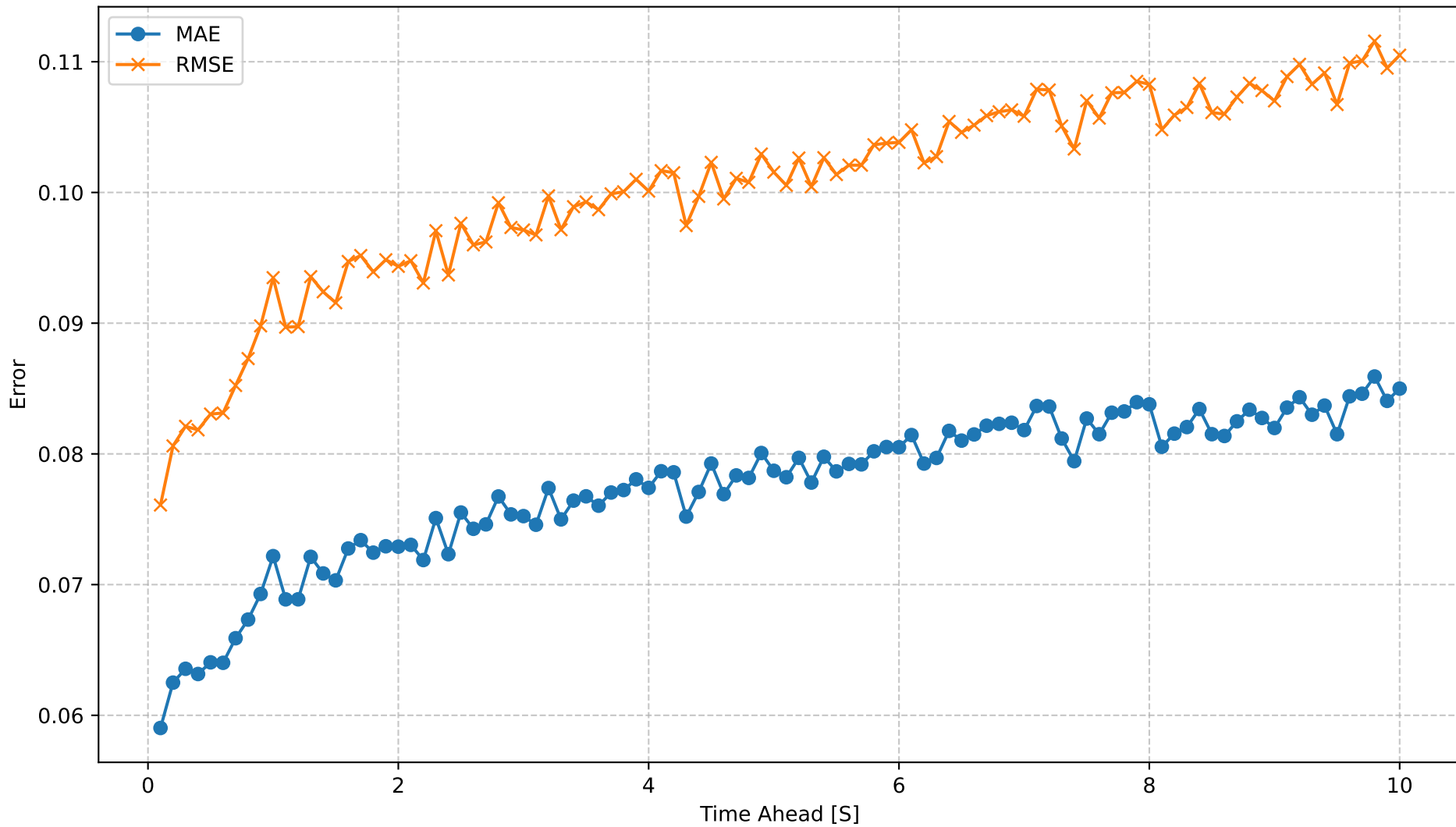
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 100 $\approx$ 10.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: Training: rssi_odom_dataset_12apr25_isolado.parquet |
Experiment: rssi_odom_dataset_11apr25_isolado.parquet
Modelo: GRU
Forecast steps: 100

MAE / RMSE (dB)

- 1º passo : 0.06 / 0.08
- Mediana : 0.08 / 0.10
- 100º passo : 0.08 / 0.11

Crescimento do erro: 44.0% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 98 (0.09), 100 (0.08), 97 (0.08)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.